

## **DYSPEPSIA**

Key Family Physician: Dr Phua Yiyong

Specialist Advisors: Dr Tang Si Ying (Consultant, Alexandra Hospital ) and Dr Alexander Yip (Consultant, Alexandra Hospital )

*Acknowledgement for their inputs: NUHSP: Ms Marissa Chin, Dr Zhang Zhi Peng, Dr Tan Wee Hian and Dr Benjamin Wong*

*Published in August 2025*

# CONTENTS

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>Introduction.....</b>                           | <b>3</b>  |
| Importance .....                                   | 3         |
| Epidemiology.....                                  | 3         |
| <b>Screening and Diagnosis .....</b>               | <b>4</b>  |
| Screening.....                                     | 4         |
| Diagnosis .....                                    | 4         |
| Initial Assessment.....                            | 4         |
| Risk .....                                         | 8         |
| <b>Management and Follow-up .....</b>              | <b>9</b>  |
| Initial Management .....                           | 9         |
| Treatment Goals / Targets of Control.....          | 13        |
| Non-Pharmacological / Lifestyle .....              | 13        |
| Pharmacological / Medications .....                | 14        |
| Treatment Algorithm for Functional Dyspepsia ..... | 15        |
| Treatment Algorithm for Reflux .....               | 16        |
| Gastrointestinal Reflux .....                      | 16        |
| Detection & Treatment of Complications.....        | 17        |
| Urgent and Emergency Situations .....              | 17        |
| Referrals to Hospital .....                        | 17        |
| <b>References .....</b>                            | <b>17</b> |

## **INTRODUCTION**

Dyspepsia or more commonly called indigestion refers to pain or discomfort in epigastric area (central upper abdomen and lower chest area) attributed to the Gastrointestinal System for a period of at least 1 month. (Moayyedi & Lacy, 2017). It typically occurs after meals but can also happen if there is no intake of food during normal mealtimes. Other associated symptoms include the following:

- Burping.
- Feeling bloated.
- Feeling nauseated.
- Heartburn (burning sensation behind sternum).
- Early satiety.

## **Importance**

Pain and discomfort in the epigastric area often poses a diagnostic challenge. It can be a symptom of digestive organs as well as organs around the area inclusive of the heart, liver, gallbladder and pancreas.

The imperative is hence to identify patient with a time sensitive disorder (e.g. myocardial injury or gastric cancer) that requires expedited and targeted tertiary investigation versus those who can be managed in primary care.

Nevertheless, a patient with dyspepsia is usually distressed and the aim of this guide is to approach this condition appropriately and safely.

## **Epidemiology**

Locally, up to 38% of Singaporeans have dyspepsia at some point or another (W Luman, 2001). It is more common in females, smokers and those who take NSAIDs. (Moayyedi & Lacy, 2017). The most common type of dyspepsia at up to 70% is “Functional Dyspepsia” where a structural cause is not found. (Chona DL, 2021)

## **SCREENING AND DIAGNOSIS**

### **Screening**

[See NUP Cancer Screening Guideline](#) (Page 16)

### **Diagnosis**

#### **Initial Assessment**

There are many causes of epigastric pain.

A PROMPT Model (Murtagh, 2024) is suggested for the primary care practitioner as a practical diagnostic tool.

**P** – Probable diagnosis. What is the most likely and common diagnosis?

**R** – Red Flags. What are the serious disorders not to be missed?

**O** – Often Missed Causes. E.g. medications

**M** – Masquerades. E.g. depression or poorly controlled DM causing gastroparesis.

**PT** – is the patient trying to tell me something?

#### **P: Probable diagnosis**

From a statistical viewpoint, most of the presentation of dyspepsia is likely due to a luminal cause (functional dyspepsia, gastritis or reflux), however due diligence must be taken to rule out other causes.

The key is to establish the likelihood of the epigastric pain presentation being more associated with the GI tract (luminal causes) or other nearby organ.

Key history taking that helps to establish the probable diagnosis mainly includes the following:

- Patient age and demographics
- Symptoms and signs.
- Onset, time course and duration
- Risk factors

#### **Cardiac causes**

A cardiac cause of epigastric pain (ischemic heart disease) is usually time sensitive and should be one of the first differentials to be ruled out. The pain or discomfort of a cardiac origin will typically be on exertion and occur at timings not related to meals. Other characteristics pointing to a cardiac origin may include the following:

- Presence of risk factors for cardiac issues such as family history of ASCVD, smoking, poorly controlled cardiovascular risk diseases, obesity and older age

- Associated symptoms such as diaphoresis and dyspnoea
- Character of pain or discomfort is usually different – ischemic in nature, e.g. tight, aching or crushing

### Gastro-intestinal tract causes

Gastrointestinal causes can be classified as **luminal** or **extra luminal causes**.

**Luminal causes** will typically (but not always) include the following:

- Onset of pain or discomfort related to meals or meal timings
- GI symptoms such as burping, feeling nauseated, bloating, early fullness (satiety) or heart burn

#### 1. Functional Dyspepsia (FD)

Dyspepsia is considered functional when no structural pathology has been identified. Studies have shown that it can account for up to 70% of patient with dyspepsia.

The ROME IV criteria highlighted 4 key symptoms to the diagnosis FD.

| Functional Dyspepsia        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diagnostics criteria</b> | <p>One or more of the following:</p> <ul style="list-style-type: none"> <li>i. Bothersome postprandial fullness</li> <li>ii. Bothersome early satiation</li> <li>iii. Bothersome epigastric pain</li> <li>iv. Bothersome epigastric burning</li> </ul> <p><b>AND</b></p> <p>No evidence of structural disease (including upper endoscopy) that is likely to explain the above symptoms.</p> <p>Criteria fulfilled for the last 3 months with symptoms onset at least 6 months prior to diagnosis.</p> |

#### 2. Gastritis

Occurs when the lining of the stomach has inflammation.

- a) *H. pylori* infection is one of the most common causes of gastritis. The prevalence of *H. pylori* in Singapore is 31% (Ang TL, 2016).
- b) Other causes can be classified as exogenous causes (NSAIDs, Steroids or alcohol over consumption) or endogenous causes (usually autoimmune causes).

### 3. Gastroesophageal Reflux Disease (GERD)

Occurs when stomach acid rises into the oesophagus.

- a) Might occur after a full meal or on lying down after a meal.
- b) Triggers might be identified – see [Non-Pharmacological Management](#)
- c) Associated symptoms might include the following.
  - i. Heartburn (burning pain behind the sternum), possibly rising up the throat.
  - ii. Backwash of food or sour taste in the throat.
  - iii. Sensation of a lump in the throat or cough after lying down.

#### **Extra-luminal gastrointestinal causes**

##### 1. Biliary Colic

Colic is pain experienced when there is a buildup of pressure in a hollow organ. Biliary colic occurs when the biliary tree is obstructed, usually due to passage of gallstones.

- a) Might occur after a large or fatty or oily meal (e.g. chicken rice)
- b) Pain is usually sharp / cramping / squeezing and quickly escalating before resolving as pressure is relieved when the stone passed
- c) Pain might radiate to the upper back or right shoulder
- d) Ultrasound Hepato-biliary system usually reveals gallstones

##### 2. Chronic Pancreatitis

Pancreatitis is inflammation of the pancreas. Patient with chronic pancreatitis usually have the following.

- a) Epigastric pain that might radiate to the back and associated with nausea and vomiting
- b) Endocrine insufficiency resulting in impaired glucose metabolism or diabetes mellitus
- c) Weight loss, chronic diarrhoea or steatorrhea as a result of malabsorption from exocrine insufficiency

The most common cause of chronic pancreatitis is alcohol overuse or abuse, followed by pancreatic duct damage from acute pancreatitis and obstructive causes.

**R: Red Flags** for disorders not to be missed in primary care

Cardiac Red Flags:

- New ECG changes (e.g. Q-waves, TWIs *etc.*)
- New reduction in effort tolerance
- New onset - Exertional chest pain

Gastrointestinal Red Flags as below:

Malignancy (gastric or pancreatic tumour)

- New onset persistent dyspepsia above the age of 45
- Unintentional, significant weight loss (at least 5% over 6 to 12 months)
- Family history of gastric cancer
- Dysphagia (Difficulty in swallowing)
- Odynophagia (Painful when swallowing)
- Epigastric or abdominal mass
- Melena or haematemesis
- Risk factors for Barrett's Oesophagus
  - Chronic GERD of at least 5 years (persistent symptoms) + 3 of the additional risk factors below:
    - Age > 50
    - Male
    - Obesity (BMI > 23.5 kg/m<sup>2</sup>)
    - Past or active tobacco use
    - 1<sup>st</sup> degree relative with Barrett's oesophagus or oesophageal adenocarcinoma

Peptic Ulcers Disease:

- Melena
- Haematemesis

Pancreatitis or cholecystitis:

- Unwell looking
- Fever
- Tender abdomen, positive Murphy's finding

## **O: Often Missed**

### Medication

Typically, there will be a temporal relationship to the onset of the epigastric pain and initiation of the offending medication.

The medications that can cause epigastric discomfort more commonly seen in primary care are listed below.

- Diabetic medication such as metformin, acarbose & GLP-1 agonist.
- Iron tablets
- NSAIDs
- Antidepressants (SSRI's)
- Potassium chloride
- Orlistat
- Antibiotics especially doxycycline
- Herbs (Ginkgo for cognition, Saw palmetto for BPH, garlic).

### Alcohol Gastritis

Alcohol is known to irritate the stomach lining. Long term or acute large consumption of alcohol can erode the lining leading to inflammation and dyspepsia-like symptoms.

## **M: Masquerades**

Depression / Anxiety – patient with depression or anxiety can sometimes have somatization of their mental health disorders as complaints of epigastric pain.

Aerophagy – occurs when a patient swallowed too much air during rapid eating or drinking or sub-consciously as a habit.

Oesophageal Spasm and Biliary Motility Disorder are other more rare causes of dyspepsia. They are usually diagnosed by a Gastroenterologist.

## **OT: Other things**

Is the patient trying to tell me something? Explore the patient ideas, concerns, and expectations.

## **Risk Stratification**

See Red Flags.

## **MANAGEMENT AND FOLLOW-UP**

### **Initial Management**

The general approach for dyspepsia is to exclude non-luminal cause of dyspepsia and red flags first and foremost. If red flags are present, an urgent or early referral to Gastroenterology or another appropriate speciality is warranted. Non luminal causes of dyspepsia should be referred appropriately, e.g. biliary colic to Hepatobiliary Surgery.

After ruling out red flags and extra-luminal causes of dyspepsia, a test and treat approach (Chew CA, 2017) for luminal causes of **persistent**<sup>1</sup> dyspepsia should be adopted.

The investigation available in NUP are listed below.

- 1) Open Access OesophagoGastroDuodenoscopy (OGD)
- 2) Open Access UBT (pending availability)
- 3) GastroClear

The choice of investigation should be based on a patient centric approach bearing in mind the below:

Features of each diagnostic test:

|                        | Open Access UBT                                                                                                                                                                                                                                         | Open Access OGD                                                 | GastroClear                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Age limitations</b> | Above 16 years of age                                                                                                                                                                                                                                   | Between 21 and 60 years of age                                  | Above 40 years of age                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Indications</b>     | Positive <i>H. pylori</i> Antibodies on blood test <sup>2</sup> .<br><br>Dyspepsia symptoms without red flags (see above)<br><br>Close family contact of family members recently diagnosed with <i>H. pylori</i> TRO similar <i>H. pylori</i> infection | Recurrent upper abdominal pain / bloating<br>Reflux / heartburn | As a screening test to identify intermediate to high-risk patients who should undergo gastroscopy for more detailed examination to detect gastric cancer.<br>Symptomatic patients without red flags for dyspepsia who are not keen for gastroscopy can consider GastroClear as first line investigation to delineate gastric cancer risk and then persuade patient to go for a gastroscopy |

<sup>1</sup> There is no universal consensus on the duration of symptoms that is considered “persistent”. In medical literature, symptoms are considered persistent if they last for more than 2 months. Physician to contextualize treatment plan from a patient centric perspective.

<sup>2</sup> *H. pylori* antibodies are indicative of either an active or past *H. pylori* infection. They will remain positive even after *H. pylori* has been eradicated and **do not** confer protection against future *H. pylori* infection.

|                            | Open Access UBT                                                                                                                                                                                                                                                                                                                                                                                                                      | Open Access OGD                                                                                                                                                                                                                                                                                                                            | GastroClear                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exclusion criteria</b>  | Age < 16.<br>Red flags present<br>Pregnant.                                                                                                                                                                                                                                                                                                                                                                                          | Refer to exhaustive <a href="#">exclusion criteria here</a><br>Age < 21 or > 60.                                                                                                                                                                                                                                                           | Age < 40.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Description of test</b> | Swallow a pill or liquid containing Urea.<br>Blow into a solution subsequently.<br><br>Detects the presence of active <i>H. pylori</i> infection which is a common cause of luminal dyspepsia.                                                                                                                                                                                                                                       | A tube containing a camera is inserted via the mouth into the GI canal under sedation.<br><br>Direct visualization of GI lumen and hence able to pick up structural abnormalities visually, via histology and detect <i>H. pylori</i> infection.<br><br>Can carry out therapeutic procedures.                                              | Venous blood test is taken.<br><br>Analyzes specific microRNA biomarkers in the blood to identify individuals at risk for gastric cancer even before symptoms appear.                                                                                                                                                                                                                                                                       |
| <b>Usage scenario</b>      | Suitable for patients with positive <i>H. pylori</i> antibodies on health screen test.<br><br>Suitable for patients who have close contact family members recently diagnosed with <i>H. pylori</i> infection TRO infection.<br><br>Suitable for younger patients with dyspepsia that do not have red flags.<br><br>Not suitable for patients with purely GERD symptoms as <i>H. pylori</i> is not a risk factor for developing GERD. | Suitable to investigate for various luminal causes of dyspepsia such as GERD symptoms, gastritis, <i>H. pylori</i> infection and functional dyspepsia.<br><br>Can consider as main modality of investigation in older patient (age > 45) with new onset of persistent dyspepsia as risk of gastric cancer increase exponentially with age. | Suitable for patients with concerns and risk factors for gastric cancer<br><ul style="list-style-type: none"> <li>- Age &gt; 40,</li> <li>- smoker,</li> <li>- history of <i>H. pylori</i> infection</li> <li>- gastric inflammation &gt; 2 years since last OGD</li> </ul> See <a href="#">information sheet</a> .<br><br>Can be offered in conjunction with UBT or OA-OGD if patient has high level of Gastric Ca concerns <sup>2</sup> . |

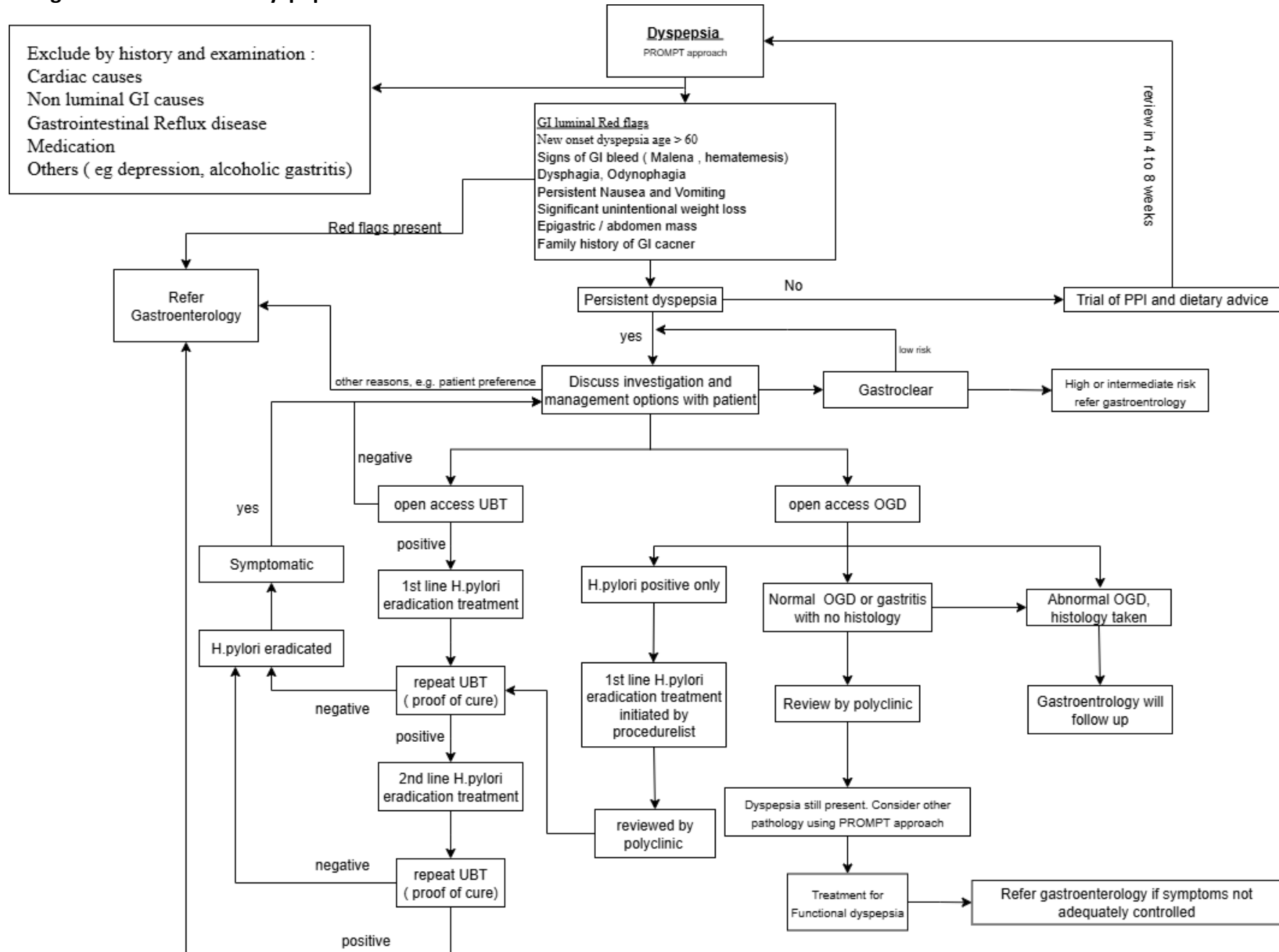
<sup>2</sup> GastroClear measures microRNA biomarkers and can possibly pick up early gastric cancer that is not seen on OGD. It can be complementary to OGD TRO gastric cancer. However, GastroClear results should reviewed first before deciding to refer low risk patient for OA-OGD or intermediate / high risk patient to gastroenterology.

|                            | Open Access UBT                                                                                                                                                                                                                                                                                                               | Open Access OGD                                                                                                                                                                                             | GastroClear                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Common Pitfalls</b>     | <p>Need to explain to patient a positive UBT requires another repeat UBT for proof of cure.</p> <p>Need to explain to patient limited diagnostic ability of test – can only detect for <i>H. pylori</i>. A negative test result might require a OGD subsequently to investigate for other causes of persistent dyspepsia.</p> | <p>Do not offer with open access UBT as there will be duplicity.</p> <p>Open Access OGD is not suitable for screening of gastric cancer in asymptomatic patient. <a href="#">See inclusion criteria</a></p> | <p>Need to explain GastroClear only stratify the risk of gastric cancer into low, intermediate or high as it is a <u>screening</u> test.</p> <p>Need to counsel that GastroClear results that are intermediate and high risk patient has to be referred to Gastroenterology for further investigation (<b>not suitable</b> for Open Access OGD which is only for low risk patient)</p> <ul style="list-style-type: none"> <li>- clinicians should defer OA OGD or referral until GastroClear result is out if there is intention to offer both OA OGD and GastroClear.</li> </ul> |
| <b>Risk</b>                | Non-Invasive and very low risk.                                                                                                                                                                                                                                                                                               | invasive procedure ( e.g. GI tract bleeding, perforation) and sedation related risk.                                                                                                                        | <del>Low risk invasive blood taking procedure – risk of pain and infection.</del> Low Risk – venipuncture related risk.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Diagnostic accuracy</b> | Test has high sensitivity (94.2%) and specificity (100%) for <i>H. pylori</i> (T S Chua, 2000)                                                                                                                                                                                                                                | Operator dependent test - the procedurist has varying threshold and experience.                                                                                                                             | Overall Sensitivity = 87% Specificity = 68%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Cost</b>                | About \$150 per test for a Singapore Citizen                                                                                                                                                                                                                                                                                  | At least \$700 for a Singapore Citizen (additional cost for histology, therapeutic procedure)                                                                                                               | About \$80 till further notice.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Other practice tips:

- Refer to the [Open Access OGD protocol](#) for inclusion criteria for suitability. Patient who fulfils exclusion criteria can be referred to Gastroenterology for further management.
- Refer [Figure 1](#) on evaluation of dyspepsia algorithm.

**Figure 1: Evaluation of Dyspepsia**



## Treatment Goals / Targets of Control

Goals of treatment is to restore patient function and be symptom free.

## Non-Pharmacological / Lifestyle

Certain triggers or modulators can be identified to trigger functional dyspepsia or reflux symptoms. Identifying and correcting them can help patient manage their symptoms.

Adapted from (Kahrilas PJ, 2024)

| Categories                              | Trigger / Modulators                                                                                                      | Suggested lifestyle and diet changes                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eating behaviour</b>                 | Eating too fast.<br>Large portions.<br>Eating beyond satiety.<br>Eating close to bedtime.<br>Irregular mealtimes.         | Serve smaller portions to reduce meal volume.<br>Use smaller plates to feel satisfied with smaller amount.<br>Finish eating 3 hours before lying down.<br>Schedule proper meals to avoid grazing.                                                                                                                                                                            |
| <b>Dietary factors</b>                  | Fatty / Fried / Oily food.<br>Spicy or sour food.<br>Excessive alcohol.<br>Excessive coffee or tea.<br>Carbonated drinks. | Add in fruits and vegetable into each meal via healthy plate, however, avoid citrus.<br>Reduce fatty / fried / oily food. Replace meat base protein with plant based e.g. tofu.<br>Eliminate or wean down caffeine intake.<br>Limit alcohol. No alcohol before bed.<br>Choose water over sweetened carbonated drinks.                                                        |
| <b>Lifestyle</b>                        | Weight gain.<br>Smoking.<br>Tight garments or belt.                                                                       | Lose weight. Incorporate activity into lifestyle, e.g. aim 7000 to 10000 steps per day. Accumulate 150 minutes of moderate physical exercise per week.<br>Stop smoking.<br>Wear loose clothing to reduce pressure around abdomen.<br>Lie on left side to minimise reflux. Elevate head of the bed when sleeping, e.g. with wedge form to form an inclined sleeping position. |
| <b>Emotional or behavioural factors</b> | High stress environment or period.<br>Hypervigilance.                                                                     | Practice deep breathing or other stress releasing techniques.                                                                                                                                                                                                                                                                                                                |

## Pharmacological / Medications

The 3 main luminal causes of dyspepsia that can be managed in primary care include *H. pylori* infection, functional dyspepsia and GERD.

### *H. pylori* Infection

*H. pylori* can be detected via either Open Access UBT or Open Access OGD.

Under the Open Access OGD protocol, the Gastroenterologist will initiate 1<sup>st</sup> line *H. pylori* eradication treatment. Patient will return to polyclinic for review and proof of cure with a UBT under the Open Access OGD protocol. Refer to [Figure 1](#) for further clarity.

If *H. pylori* is detected via Open Access UBT, NUP doctor has to initiate 1<sup>st</sup> line *H. pylori* eradication treatment. Refer to [Figure 1](#) for further clarity.

1. 1<sup>st</sup> line *H. pylori* treatment eradication treatment
  - i. Omeprazole 20 mg BD, amoxicillin 1 gram BD, clarithromycin 500mg BD for 2 weeks (1<sup>st</sup> line) if **no penicillin allergy**
  - ii. Omeprazole 20mg BD, metronidazole 400mg TDS, clarithromycin 500mg BD for 2 weeks if **there is penicillin allergy**.
2. 2<sup>nd</sup> line *H. pylori* eradication treatment
  - i. Omeprazole 20mg BD bismuth subcitrate 240mg BD / bismuth subsalicylates 525mg QDS<sup>3</sup>, metronidazole 400mg TDS, tetracycline<sup>4</sup> 500mg QDS for 2 weeks.

Refer to this [resource](#) for latest updated treatment for *H. pylori* infection.

---

<sup>3</sup> Not stocked in NUP. To provide an external prescription. NUH / AH carry bismuth subsalicylate, NTFGH carries bismuth subcitrate

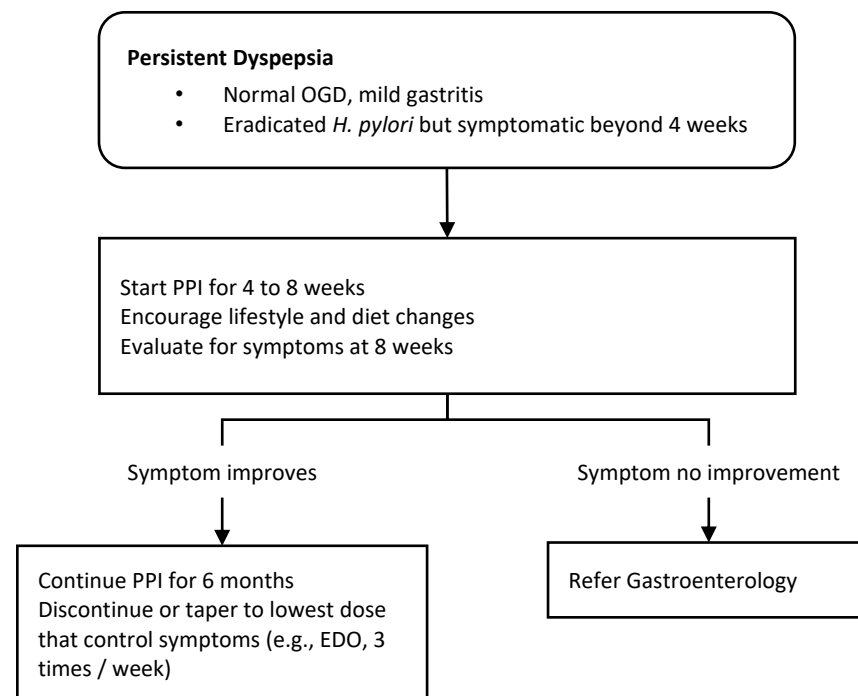
<sup>4</sup> Not in stock in NUP, to provide external prescription

Patients with *H. pylori* that remained symptomatic after 4 weeks after proven eradication can be treated following the functional dyspepsia pathway as below.

### Treatment Algorithm for Functional Dyspepsia, adapted and modified from (Uptodate, 2025)

#### Functional Dyspepsia

1. Proton pump inhibitors remain the mainstay of initial treatment and is likely more efficacious compared to H2 Receptor Antagonist (H2RA)
  - Omeprazole 20mg OM is the recommended dose to start patient on a PPI treatment. There appears to be no added benefits for BD or higher dosage.
  - H2RA such as famotidine 10mg OM or BD is an alternative for patient who cannot tolerate or prefer not to use PPIs.
2. Initiating appropriate lifestyle and dietary changes can augment medication treatment for functional dyspepsia.
3. Persistent dyspepsia despite a trial of PPI can be referred to Gastroenterology for further management.



## Gastrointestinal Reflux

In the absence of red flags or risk factors for Barrett's esophagus<sup>1</sup>, empirical treatment of reflux can be initiated without the need to visualize the GI lumen.

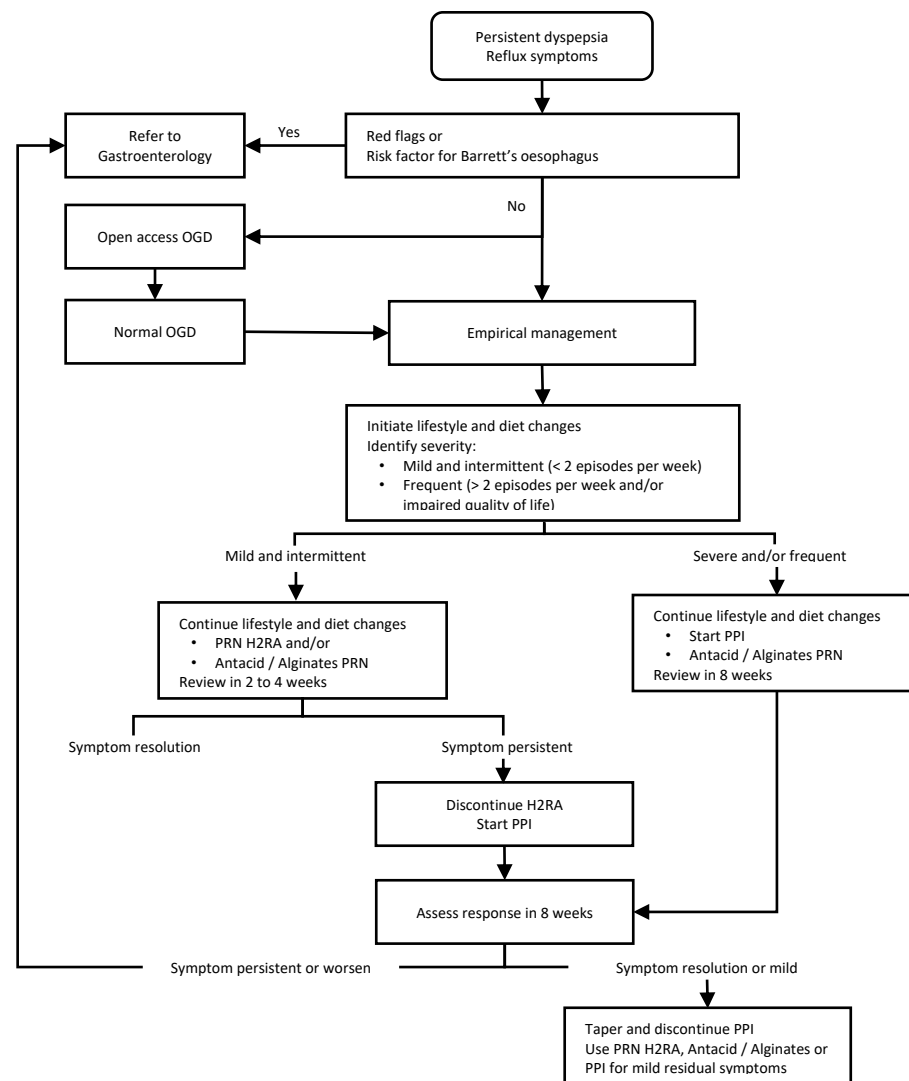
- Identified triggers and initiate diet and lifestyle changes – see non-pharmacological management.
- Identify severity to guide pharmacological treatment.
  - Mild and intermittent: less than 2 reflux episodes / week
  - Severe: 2 or more reflux episodes/ week or impaired quality of life

|                  | Common medication and dose                                                            | Remarks                                                                                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H2RA</b>      | Famotidine 20mg OD-BD.                                                                | Faster onset of action than PPI but might achieve tachyphylaxis in 4-6 weeks, hence preferentially use in mild and intermittent case.                                                                       |
| <b>PPI</b>       | Omeprazole 20mg OM.                                                                   | Requires 3 to 4 days for full onset of acid reduction effect but more efficacious than H2RA. Given for a period of 8 weeks.                                                                                 |
| <b>Antacid</b>   | Magnesium Trisilicate Mixture 5-15ml QDS PRN.<br>Antacid tablets 1-2 tablets TDS PRN. | Neutralize gastric acid.                                                                                                                                                                                    |
| <b>Alginates</b> | Gaviscon 10-20 ml TDS PRN.                                                            | Gaviscon contains carbonates to neutralize gastric acid. Additionally, the alginates which is a polysaccharide forms a viscous gel gum floating at the gastroesophageal junction to neutralize acid pocket. |

<sup>1</sup>Risk factors for Barrett's: Chronic GERD of at least 5 years persistent symptoms + 3 of the additional risk factors below:

- Age > 50
- Male
- Obesity (BMI > 23.5 kg/m<sup>2</sup>)
- Past or active tobacco use
- 1<sup>st</sup> degree relative with Barrett's oesophagus or oesophageal adenocarcinoma

## Treatment Algorithm for Reflux adapted from (Uptodate, 2025)



## Detection & Treatment of Complications

Refer to initial evaluation.

## Urgent and Emergency Situations

Refer to initial evaluation.

## Referrals to Hospital

Refer to management.

## REFERENCES

1. Ang TL, F. K. (2016). The changing profile of *Helicobacter pylori* antibiotic resistance in Singapore: a 15 year study. *Helicobacter*, 21:261-5.
2. Chew CA, L. T. (2017). The diagnosis and management of *H. pylori* infection in Singapore. *Singapore Med J*, May;58(5):234-240.
3. Chona DL, T. M. (2021). Dyspepsia: A stepwise approach to evaluation and management. *J Fam Pract*, Sep;70(7):320-325.
4. Kahrilas PJ, A. F.-P. (2024). Assessment and treatment of reflux-like symptoms in the community: a multidisciplinary perspective. *Br J Gen Pract*, Apr 25;74(742):232-235.
5. Moayyedi, P. M., & Lacy, B. E. (2017). *ACG and CAG Clinical Guideline: Management of Dyspepsia*. The American Journal of Gastroenterology.
6. Murtagh. (2024, 07 04). *RACGP AKT KFP*. Retrieved from <https://racgp-akt-kfp.com/murtagh-prompt-model/>
7. T S Chua, K. M. (2000). Validation of 13C-Urea Breath Test for the Diagnosis of *Helicobacter pylori* Infection in the Singapore Population. *Singapore Med J*, Vol 43(8) : 408-411.
8. *Uptodate*. (2025). Retrieved from [https://www.uptodate.com/contents/image?imageKey=PC%2F146999&topicKey=PC%2F19&search=functional+dyspepsia+treatment&rank=1%7E150&source=see\\_link](https://www.uptodate.com/contents/image?imageKey=PC%2F146999&topicKey=PC%2F19&search=functional+dyspepsia+treatment&rank=1%7E150&source=see_link)
9. W Luman, H. N. (2001). Survey of Dyspepsia Management in Community. *Singapore Med J*, Vol 42(1) : 026-02.